Vamsi Muppana

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Professional Experience

Software Engineer **Dec 2024 – Present**

EarthQuest

- Architected a stateful game server layer in **Node.js** and **WebSockets** with Redis pub/sub fan-out, sustaining 15K concurrent players at p95 latency under 80ms
- Built real-time leaderboards on **Redis** sorted sets with write-behind batching to PostgreSQL, cutting database write load by 60% during spike-driven traffic events
- Shipped a live spectator mode streaming **HLS** through AWS IVS to React and React Native clients, syncing in-game events to playback using timed metadata cues
- Integrated the **Claude API** into NPC dialogue with a RAG pipeline over pgvector-indexed game lore, streaming tokens and cutting first-token latency 45% via prompt caching
- Developed cross-platform game UI and in-game messaging in **React**, TypeScript, and React Native with Zustand state, holding 60 FPS interactions on mid-range mobile devices
- Drove AI-native delivery using **Claude Code** and Cursor agents for scaffolding, refactors, and test generation, raising unit test coverage to 85% across core game services
- Owned features end to end on AWS ECS with Docker and GitHub Actions, running k6 load tests at 10x burst traffic before releases and leading production incident follow-ups

Software Engineer **Jun 2022 – Jul 2024**

Skyline Infrastructure

- Built REST and GraphQL APIs in **Node.js** and TypeScript for a customer-facing service platform, handling 60K+ daily requests with p95 response times under 150ms
- Developed a real-time notification service with **Socket.IO** and Redis pub/sub, pushing live status updates to 20K+ users and replacing 30-second polling on web clients
- Implemented Kafka-based event processing in **Java Spring Boot** to decouple order updates from billing, enabling independent scaling and idempotent retries on failure
- Engineered React dashboards with virtualized lists and memoized components to render 10K+ live records smoothly, cutting initial load time 40% through code splitting
- Added Redis caching with TTL-based invalidation for read-heavy endpoints, lowering database pressure during peak hours and improving API throughput by 2.5x
- Containerized services with **Docker** and built GitHub Actions CI/CD pipelines with automated tests and staged AWS rollouts, reducing deployment time from 45 to 12 minutes
- Wrote unit and integration tests with Jest and Supertest to reach 80% coverage, and added structured logging with Datadog dashboards to speed up production incident triage

Projects

AI-Powered Social Media Scheduling Tool | *Next.js, Node.js, Claude API, PostgreSQL, Redis, WebSockets* **2025**

- Built a creator-focused scheduler where the **Claude API** drafts platform-specific captions and hooks, with BullMQ job queues publishing posts on time across 4 platforms
- Streamed live engagement metrics to a Next.js dashboard over **WebSockets**, using Redis-backed aggregation to update counters in real time without polling the database

Technical Skills

Languages: JavaScript, TypeScript, Java, Python, SQL, HTML, CSS

Frameworks & Libraries: React, React Native, Next.js, Node.js, Express, Socket.IO, Spring Boot, GraphQL

Databases: PostgreSQL, Redis, MongoDB, DynamoDB, pgvector

Cloud & DevOps: AWS (ECS, Lambda, S3), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD

Tools & Practices: Claude Code, Cursor, AI Coding Agents, Claude API, AI-Assisted QA, Playwright, Jest, Datadog, Git, Agile

Core Concepts: Real-Time Systems, WebSockets, Live Streaming (HLS, RTMP), Leaderboards, Gaming Backend Services, Distributed Systems, Load Balancing, Microservices, System Design

Education

University at Albany, SUNY, Albany, NY – Master's in Computer Science (AI/ML), GPA: 3.68/4.0
Gayatri Vidya Parishad College of Engineering, India – Bachelor's in Computer Science

Certifications

Oracle Certified Generative AI Professional | Oracle Certified Data Science Professional